Generating Chest Radiographs with a Convolutional Variational Autoencoder

ECE 285 - Homework 1

Abstract

A convolutional variational autoencoder (VAE) was trained on pediatric chest radiographs from the Kaggle pneumonia dataset. We report the reconstruction and KL losses, unconditional prior samples, and Ignite Inception Score (IS) and Fréchet Inception Distance (FID) against held-out test radiographs. The measured scores describe image statistics only; they do not establish clinical fidelity.

1 Data and preprocessing

We used the *Chest X-Ray Images (Pneumonia)* dataset [1]. Its original training split was divided by class into 4694 training and 522 validation images with seed 285. The separate test split (624 images) was reserved for generative evaluation. We converted each image to one grayscale channel, resized it to 64 x 64 pixels, and scaled values to [0, 1].

2 VAE and objective

Four stride-2 convolutional layers map an image to 4 x 4 x 128 features. Two linear heads produce the mean and log variance of a 32-dimensional Gaussian posterior. A reparameterized sample enters a linear layer and four transposed convolutions. A sigmoid decoder outputs grayscale pixels. The model has 543,649 trainable parameters [2].

We minimized pixelwise binary cross entropy summed over all pixels plus beta times KL(q(z|x) || N(0,I)), averaged per image. The KL term is analytic; samples use the reparameterization trick. For generation, z is drawn from the unit Gaussian prior, without using a test image.

3 Training configuration

Setting	Value
Image / latent	64 x 64 / 32
Batch / epochs	64 / 20
Optimizer / learning rate	Adam / 0.001
KL weight beta	1
Device / seed	cpu / 285

Checkpoint selection used the lowest validation negative beta-ELBO. The official test data were never used to update weights or choose the checkpoint.

4 Training behavior

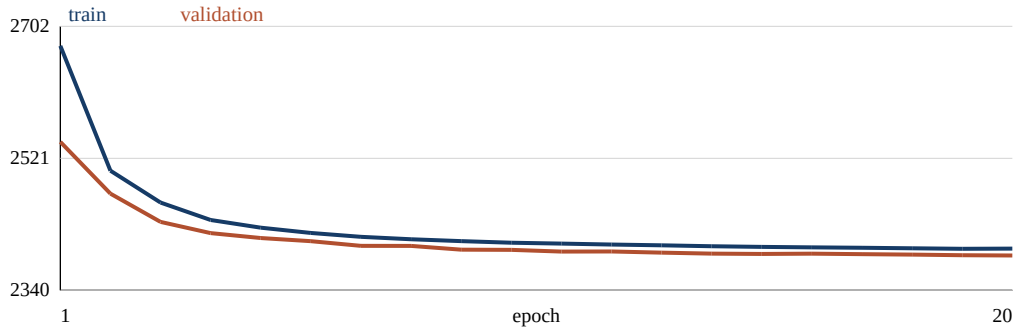


Figure 1: mean negative beta-ELBO (nats/image) on the training and validation partitions; lower is better.

The selected checkpoint was epoch 20, with validation loss 2388.1 (BCE 2366.0, KL 22.1) nats/image. The final training loss was 2397.5.

5 Generative evaluation

We generated 256 independent prior samples and compared them with 256 randomly selected official test images. For both metrics, grayscale images were copied into three channels, resized to 299 x 299, and processed by ImageNet-pretrained Inception-v3 through PyTorch-Ignite [3].

Metric	Measured result
Inception Score	1.779
FID-1000	0.177

FID-1000 uses the 1000 ImageNet class probabilities as features, matching the Ignite tutorial. It is not directly comparable to pool3/2048 FID. IS and FID are sensitive to sample count and natural-image classifier bias; neither measures diagnostic validity [3,4].

6 Qualitative results and discussion

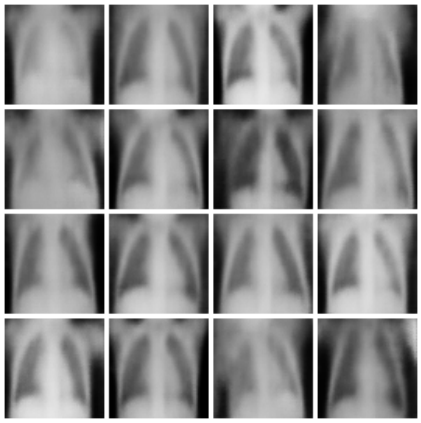


Figure 2: generated samples

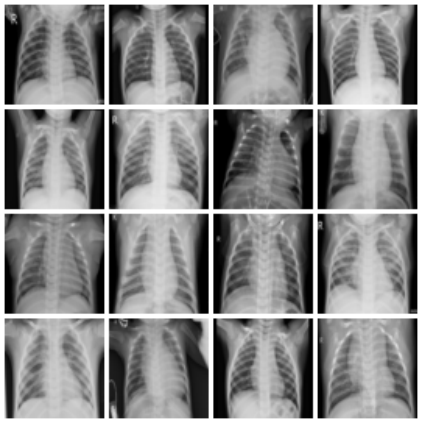


Figure 3: held-out test images

Both grids use the same 64 x 64 grayscale scale. The generated samples recover coarse chest shape but lose much of the rib and lung detail visible in held-out scans. Generation was unconditional; pathology cannot be selected or inferred.

7 Limitations

The 64 x 64 resolution discards fine lung detail. Pixelwise BCE can favor smooth averages, while the unit Gaussian prior and a single decoder may not capture all acquisition and anatomical variation. The training set is class-imbalanced, and this unconditional VAE does not control pathology.

The reported scores are descriptive for this exact 256-sample experiment. Inception-v3 was trained on natural photographs, so its class probabilities are a weak proxy for radiographic structure. FID-1000's feature choice should accompany every comparison. A clinician review or a validated radiology-specific feature model would be needed for stronger medical conclusions.

8 Conclusion

A compact convolutional VAE can be trained end to end on this dataset and sampled from its learned prior. The loss curve, paired examples, IS, and FID-1000 provide a reproducible baseline. Further work should test higher resolution, conditional generation, and radiology-aware evaluation.

References

[1] P. Mooney. Chest X-Ray Images (Pneumonia). Kaggle, 2018. https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia

[2] D. P. Kingma and M. Welling. Auto-Encoding Variational Bayes. ICLR, 2014.

[3] PyTorch-Ignite. GAN evaluation using FID and IS. https://pytorch-ignite.ai/blog/gan-evaluation-with-fid-and-is/

[4] M. Heusel et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. NeurIPS, 2017.